

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.51)

Metric

$$ds^2 = -dt \otimes dt + \left(\frac{F(t)^2}{\epsilon r^2 + 1} \right) dr \otimes dr + V(t, z, r, \phi)^2 dz \otimes dz + r^2 F(t)^2 d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, z, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} \right) dt + \left(\frac{\sqrt{\frac{1}{2}} F(t)}{\sqrt{\epsilon r^2 + 1}} \right) dr$$

$$n = \left(\sqrt{\frac{1}{2}} \right) dt + \left(-\frac{\sqrt{\frac{1}{2}} F(t)}{\sqrt{\epsilon r^2 + 1}} \right) dr$$

$$m = \left(\sqrt{\frac{1}{2}} V(t, z, r, \phi) \right) dz + \left(i \sqrt{\frac{1}{2}} r F(t) \right) d\phi$$

$$\bar{m} = \left(\sqrt{\frac{1}{2}} V(t, z, r, \phi) \right) dz + \left(-i \sqrt{\frac{1}{2}} r F(t) \right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(\frac{\sqrt{2}\sqrt{\epsilon r^2 + 1}}{2F(t)} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} \right) dt + \left(-\frac{\sqrt{2}\sqrt{\epsilon r^2 + 1}}{2F(t)} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2V(t, z, r, \phi)} \right) dz + \left(\frac{i\sqrt{2}}{2rF(t)} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2V(t, z, r, \phi)} \right) dz + \left(-\frac{i\sqrt{2}}{2rF(t)} \right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} \partial_t + \frac{\sqrt{2}\sqrt{\epsilon r^2 + 1}}{2F(t)} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} \partial_t - \frac{\sqrt{2}\sqrt{\epsilon r^2 + 1}}{2F(t)} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2V(t, z, r, \phi)} \partial_z + \frac{i\sqrt{2}}{2rF(t)} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2V(t, z, r, \phi)} \partial_z - \frac{i\sqrt{2}}{2rF(t)} \partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}\epsilon r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{\epsilon r^2 + 1}F(t)V(t, z, r, \phi)} - \frac{\sqrt{2}\epsilon r}{4\sqrt{\epsilon r^2 + 1}F(t)} + \frac{\sqrt{2}\frac{\partial}{\partial t} F(t)}{4F(t)} + \frac{\sqrt{2}\frac{\partial}{\partial t} V(t, z, r, \phi)}{4V(t, z, r, \phi)} - \frac{\sqrt{2}\frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{\epsilon r^2 + 1}F(t)V(t, z, r, \phi)} - \frac{\sqrt{2}}{4\sqrt{\epsilon r^2 + 1}rF(t)}$$

$$\sigma = -\frac{\sqrt{2}\epsilon r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{\epsilon r^2 + 1} F(t) V(t, z, r, \phi)} + \frac{\sqrt{2}\epsilon r}{4\sqrt{\epsilon r^2 + 1} F(t)} - \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)} + \frac{\sqrt{2} \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\sqrt{2} \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{\epsilon r^2 + 1} F(t) V(t, z, r, \phi)} + \frac{\sqrt{2}}{4\sqrt{\epsilon r^2 + 1} r F(t)}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\epsilon r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{\epsilon r^2 + 1} F(t) V(t, z, r, \phi)} - \frac{\sqrt{2}\epsilon r}{4\sqrt{\epsilon r^2 + 1} F(t)} - \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)} - \frac{\sqrt{2} \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\sqrt{2} \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{\epsilon r^2 + 1} F(t) V(t, z, r, \phi)} - \frac{\sqrt{2}}{4\sqrt{\epsilon r^2 + 1} r F(t)}$$

$$\lambda = -\frac{\sqrt{2}\epsilon r^2 \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{\epsilon r^2 + 1} F(t) V(t, z, r, \phi)} + \frac{\sqrt{2}\epsilon r}{4\sqrt{\epsilon r^2 + 1} F(t)} + \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)} - \frac{\sqrt{2} \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\sqrt{2} \frac{\partial}{\partial r} V(t, z, r, \phi)}{4\sqrt{\epsilon r^2 + 1} F(t) V(t, z, r, \phi)} + \frac{\sqrt{2}}{4\sqrt{\epsilon r^2 + 1} r F(t)}$$

$$\nu = 0$$

$$\alpha = \frac{i\sqrt{2} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t) V(t, z, r, \phi)}$$

$$\beta = \frac{i\sqrt{2} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t) V(t, z, r, \phi)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)}$$

$$\gamma = \frac{\sqrt{2} \frac{\partial}{\partial t} F(t)}{4 F(t)}$$

Ricci Tensor Components

$$\begin{aligned} \Phi_{00} = & -\frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} \\ & + \frac{\sqrt{\epsilon r^2 + 1} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 F(t) V(t, z, r, \phi)} - \frac{\sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi)} - \frac{\epsilon}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} \end{aligned}$$

$$\Phi_{01} = \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4rF(t)V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4rF(t)^2 V(t, z, r, \phi)} - \frac{i \sqrt{\epsilon r^2 + 1} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4rF(t)^2 V(t, z, r, \phi)} + \frac{i \sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4r^2 F(t)^2 V(t, z, r, \phi)}$$

$$\Phi_{02} = -\frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} + \frac{\epsilon}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}$$

$$\Phi_{10} = -\frac{i}{4rF(t)V(t,z,r,\phi)}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi) + \frac{i}{4rF(t)^2V(t,z,r,\phi)}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi) + \frac{i\sqrt{\epsilon r^2+1}}{4rF(t)^2V(t,z,r,\phi)}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) - \frac{i\sqrt{\epsilon r^2+1}}{4r^2F(t)^2V(t,z,r,\phi)}\frac{\partial}{\partial\phi}V(t,z,r,\phi)$$

$$\Phi_{11} = -\frac{\frac{\partial^2}{(\partial t)^2}F(t)}{4F(t)} + \frac{\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial t}V(t,z,r,\phi)}{4F(t)V(t,z,r,\phi)} - \frac{\epsilon r\frac{\partial}{\partial r}V(t,z,r,\phi)}{4F(t)^2V(t,z,r,\phi)} - \frac{\frac{\partial}{\partial r}V(t,z,r,\phi)}{4rF(t)^2V(t,z,r,\phi)} - \frac{\frac{\partial^2}{(\partial\phi)^2}V(t,z,r,\phi)}{4r^2F(t)^2V(t,z,r,\phi)}$$

$$\Phi_{12} = \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4rF(t)V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4rF(t)^2 V(t, z, r, \phi)} + \frac{i \sqrt{\epsilon r^2 + 1} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4rF(t)^2 V(t, z, r, \phi)} - \frac{i \sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4r^2 F(t)^2 V(t, z, r, \phi)}$$

$$\Phi_{20} = -\frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} + \frac{\epsilon}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}$$

$$\Phi_{21} = -\frac{i}{4rF(t)}\frac{\partial^2}{\partial t\partial\phi}V(t,z,r,\phi) + \frac{i}{4rF(t)^2}\frac{\partial}{\partial t}F(t)\frac{\partial}{\partial\phi}V(t,z,r,\phi) - \frac{i\sqrt{\epsilon r^2+1}}{4rF(t)^2}\frac{\partial^2}{\partial r\partial\phi}V(t,z,r,\phi) + \frac{i\sqrt{\epsilon r^2+1}}{4r^2F(t)^2}\frac{\partial}{\partial\phi}V(t,z,r,\phi)$$

$$\begin{aligned}\Phi_{22} = & -\frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} \\ & - \frac{\sqrt{\epsilon r^2 + 1} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 F(t) V(t, z, r, \phi)} + \frac{\sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi)} - \frac{\epsilon}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}\end{aligned}$$

$$\begin{aligned}\Lambda = & -\frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} F(t)^2}{12 F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{6 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{6 F(t) V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{12 V(t, z, r, \phi)} - \frac{\epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi)}{6 F(t)^2 V(t, z, r, \phi)} \\ & - \frac{\epsilon}{12 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{12 r F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{12 r^2 F(t)^2 V(t, z, r, \phi)}\end{aligned}$$

Weyl Tensor Components

$$\begin{aligned}\Psi_0 = & -\frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} \\ & + \frac{\sqrt{\epsilon r^2 + 1} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 F(t) V(t, z, r, \phi)} - \frac{\sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi)} + \frac{\epsilon}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}\end{aligned}$$

$$\Psi_1 = -\frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4 r F(t) V(t, z, r, \phi)} + \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)} + \frac{i \sqrt{\epsilon r^2 + 1} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)} - \frac{i \sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r^2 F(t)^2 V(t, z, r, \phi)}$$

$$\begin{aligned}\Psi_2 = & -\frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial t} F(t)^2}{12 F(t)^2} - \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{12 F(t)} - \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{12 F(t) V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{12 V(t, z, r, \phi)} \\ & + \frac{\epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} - \frac{\epsilon}{12 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{12 F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial}{\partial r} V(t, z, r, \phi)}{6 r F(t)^2 V(t, z, r, \phi)} + \frac{\frac{\partial^2}{(\partial \phi)^2} V(t, z, r, \phi)}{6 r^2 F(t)^2 V(t, z, r, \phi)}\end{aligned}$$

$$\Psi_3 = \frac{i \frac{\partial^2}{\partial t \partial \phi} V(t, z, r, \phi)}{4 r F(t) V(t, z, r, \phi)} - \frac{i \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)} + \frac{i \sqrt{\epsilon r^2 + 1} \frac{\partial^2}{\partial r \partial \phi} V(t, z, r, \phi)}{4 r F(t)^2 V(t, z, r, \phi)} - \frac{i \sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial \phi} V(t, z, r, \phi)}{4 r^2 F(t)^2 V(t, z, r, \phi)}$$

$$\begin{aligned}\Psi_4 = & -\frac{\epsilon r^2 \frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} - \frac{\frac{\partial}{\partial t} F(t)^2}{4 F(t)^2} + \frac{\frac{\partial^2}{(\partial t)^2} F(t)}{4 F(t)} + \frac{\frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} V(t, z, r, \phi)}{4 F(t) V(t, z, r, \phi)} - \frac{\frac{\partial^2}{(\partial t)^2} V(t, z, r, \phi)}{4 V(t, z, r, \phi)} - \frac{\epsilon r \frac{\partial}{\partial r} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)} \\ & - \frac{\sqrt{\epsilon r^2 + 1} \frac{\partial^2}{\partial t \partial r} V(t, z, r, \phi)}{2 F(t) V(t, z, r, \phi)} + \frac{\sqrt{\epsilon r^2 + 1} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial r} V(t, z, r, \phi)}{2 F(t)^2 V(t, z, r, \phi)} + \frac{\epsilon}{4 F(t)^2} - \frac{\frac{\partial^2}{(\partial r)^2} V(t, z, r, \phi)}{4 F(t)^2 V(t, z, r, \phi)}\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"